

AUGMENTED REALITY: Revolutionising Medical Applications



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Virtual reality (VR) and augmented reality (AR) are regarded as highly-complex combinations in the area of computer science, VR hardware and artistic vision. AR is regarded as a variation of virtual environments or VR. It is regarded as live direct or indirect view of the physical, real-world environment whose elements are augmented by computer-generated sensory inputs such as sound, video, graphics or GPS data.

By contrast, VR replaces the real world with a simulated one. With the aid of AR like addition of computer vision and object recognition, information about the surrounding world becomes more interactive and can be manipulated digitally.

Some researchers define AR as a technology that requires the use of head-mounted displays (HMDs). Any system can be regarded an AR system, provided it has the following three features:

1. It combines real and virtual environments.
2. The system is highly interactive while operating in real time.
3. It is used in a 3D environment.

The main aim of AR is to simplify your life by combining virtual information to immediate surroundings with an indirect view of the real-view environment such as live video streaming. It plays a crucial role in enhancing your perceptions and interactions with the real world.

Gene Becker of Lightning Laboratories defines AR as a technology, an area of research, a vision of future computing, an emerging commercial industry and a new medium for creative expression.

AR in medical applications

AR has continued to develop every day in



Fig. 1: Surgeons performing keyhole surgery using a head-mounted display

the last 50 years and has brought today's technology to modern landscape technology. It can be seen in various applications in military, industrial, commercial, entertainment and many more. But AR in medical technology is growing by leaps and bounds, and has revolutionised medical and healthcare industry by significant improvements and innovations. This is not only making patients' life safer but also making doctors more efficient in early detection of various diseases. It is making medical treatment more efficient and accurate.

Let us discuss the various AR technologies revolutionising the medical and healthcare industry.

Keyhole surgery. With the use of AR, various computer scientists have developed a highly-efficient computer-aided visualisation and navigation system for taking care of patients. Using this technology, a surgeon can wear an HMD before starting an operation. With the aid of a screen and display, the surgeon can look inside the patient's body, layer by layer, looking through skins, muscles and even bones. This technology can assist the surgeon to be more efficient while operating via the HMD, as the surgeon